PROJECT REPORT

ON

AI VOICE AGENT FOR HOME CREDIT

Preliminary Qualification for a Loan Against Property (LAP) Offer

Submitted in partial fulfilment of the requirements for the award of

DEPARTMENT OF COMPUTER SCIENCE / AI & ML

Submitted by

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ABSTRACT

This project report presents the design of an AI voice agent for Home Credit to contact existing customers about a pre-approved Loan Against Property (LAP) offer. The agent's purpose is to conduct a preliminary qualification conversation, collect seven eligibility checklist items, detect cases that should be routed to a loan-transfer specialist, and hand qualified leads to a senior loan expert. The design focuses on reliable business-rule execution while maintaining a natural, concise voice interaction.

The proposed conversation begins with customer verification and an availability check. If the customer is busy, the agent asks for a convenient callback time and ends the call. If the customer is available, the agent explains the offer as a benefit for an existing customer and checks whether the customer is seeking a fresh loan or help with an existing property loan or EMI reduction. Existing-loan and EMI-reduction cases follow a separate specialist route.

For fresh-loan cases, the agent collects property type, ownership status, availability of original property documents, desired loan amount, occupation and income mode, estimated property market value, and desired tenure. Explicit rules identify agricultural property, unavailable original documents, cash-mode income, and tenure outside the permitted three-to-fifteen-year range as disqualifying conditions. A requested amount above ₹75 lakh triggers an offer to proceed with ₹75 lakh rather than an automatic rejection. No minimum property market value is assumed because none is in the source requirements.

The design also includes state management, out-of-order information extraction, correction handling, interruption and diversion recovery, language configuration, retrieval-augmented context safeguards, and a testing matrix. The report defines the intended solution and test cases; actual deployed-call recordings, transcripts, and measured performance results must be added after implementation and execution.

Keywords: AI voice agent, prompt engineering, Loan Against Property, eligibility screening, conversation state, business rules, human handoff.

TABLE OF CONTENTS

Use Microsoft Word's References → Table of Contents feature to generate or update page numbers after editing. Apply Heading styles to chapter headings.

Abstract

Table of Contents

List of Figures

List of Tables

List of Abbreviations

Chapter 1: Introduction and Objectives

Chapter 2: Literature Review and Existing System

Chapter 3: Problem Statement and Requirements

Chapter 4: Proposed System and Methodology

Chapter 5: System Design and Diagrams

Chapter 6: Implementation and Testing

Chapter 7: Results and Discussion

Chapter 8: Conclusion and Future Scope

References

LIST OF FIGURES

Figure No.	Title
Figure 4.1	High-level AI voice-agent architecture
Figure 4.2	Conversation flow for fresh-loan qualification
Figure 5.1	Qualification state and decision flow
Figure 6.1	Proposed testing workflow

LIST OF TABLES

Table No.	Title
Table 2.1	Existing-flow limitations and proposed responses
Table 3.1	Functional requirements
Table 3.2	Eligibility rules
Table 5.1	Qualification state variables
Table 6.1	Test-case matrix
Table 7.1	Results evidence to be completed after testing

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
LAP	Loan Against Property
RAG	Retrieval-Augmented Generation
NLP	Natural Language Processing
EMI	Equated Monthly Instalment
USN	University Seat Number / Register Number, as applicable
UAT	User Acceptance Testing
TTS	Text-to-Speech
ASR	Automatic Speech Recognition

CHAPTER 1: INTRODUCTION AND OBJECTIVES

1.1 Introduction

Voice-based conversational systems can automate structured customer conversations while allowing people to respond in a natural language. In this assignment, an AI voice agent represents Home Credit and contacts existing customers about a pre-approved Loan Against Property (LAP) offer. The system is not intended to make a final lending decision. It gathers preliminary information, applies the stated qualification rules, and routes suitable leads to a senior loan expert.

1.2 Background

A preliminary loan conversation involves more than asking a fixed sequence of questions. Customers may provide several answers in one sentence, correct an earlier answer, interrupt the agent, ask unrelated questions, or disclose a disqualifying fact before the agent reaches the corresponding checklist item. A reliable agent therefore needs conversational understanding, memory of previously supplied information, explicit decision rules, and clear call-ending conditions.

1.3 Motivation

The project is motivated by the need to make preliminary qualification calls consistent and customer friendly. A well-designed agent can collect the same required data points across calls, avoid repeating questions, respond appropriately to diversions, and route exceptional cases without continuing the wrong workflow. It must also avoid inventing product details such as interest rates or eligibility thresholds that are not present in the approved context.

1.4 Problem Statement

Design a configurable AI voice agent that contacts an existing customer, verifies the intended person, checks availability, explains the LAP offer, identifies existing-loan or EMI-reduction intent, collects seven eligibility checklist items for fresh-loan cases, applies the specified rules, and performs a senior-expert handoff only when all required items have been answered and no disqualifying condition is present.

1.5 Objectives

Verify the intended customer before discussing the offer in detail. Respect the customer's availability and arrange a callback when the customer is busy. Present the LAP offer naturally and state that the current offer is available up to ₹75 lakh. Identify existing property loans or EMI-reduction requests and route them to a specialist flow. Capture the seven required checklist items accurately, including information supplied out of order. Apply eligibility rules consistently and stop regular flow when a disqualifying condition clears. Handle interruptions, corrections, natural speech, and diversions without losing the conversation state. Use configured identity, language, date/time, conversation history, and RAG context without fabricating information. Define a repeatable test plan covering eligible, ineligible, transfer, interruption, and edge-case scenarios.

1.6 Scope

The scope includes conversation design, a system prompt, state variables, eligibility logic, callback and specialist-routing flows, test scenarios, and evidence requirements. The source material permits implementation using a voice-agent platform such as Retell AI, Bolna, or a similar platform. The supplied reference material does not confirm that a platform has been deployed, or that real calls have been recorded.

1.7 Limitations

The design uses only the business rules specified in the assignment-preparation material. It does not establish an interest rate, processing fee, minimum property value, final credit policy, or a complete loan-underwriting decision. Those details must come from an authorized product context or a senior loan expert.

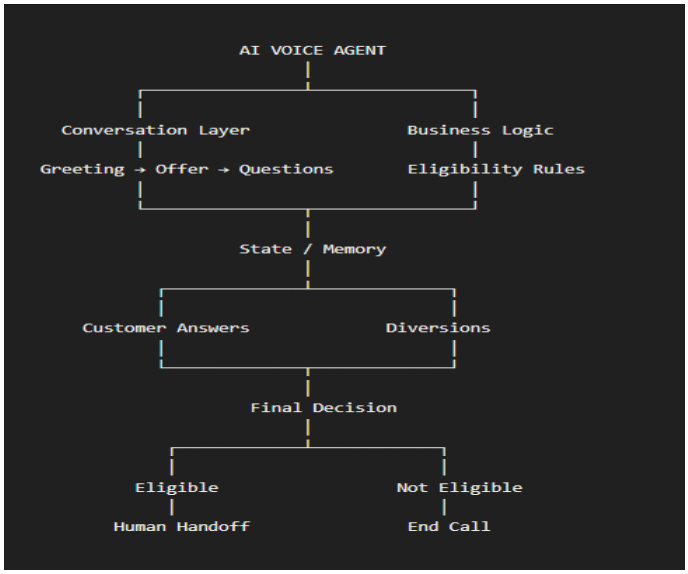
CHAPTER 2: LITERATURE REVIEW AND EXISTING SYSTEM

2.1 Basis of Review

The supplied reference file is an assignment-preparation guide rather than a collection of academic research papers. Accordingly, this chapter reviews the stated business workflow and design principles in that material. External research claims are not added. Formal research papers or product documentation can be included later if required by the institution.

2.2 Existing Workflow Described in the Requirements

The baseline workflow begins with a greeting and customer verification. The agent checks whether the customer is busy; if so, it captures a callback time and ends the call. Otherwise, it explains the LAP offers and checks for existing property-loan or EMI-reduction intent. Fresh-loan cases proceed through seven checklist items, followed by an eligibility decision. A customer who passes all checks is handed to a senior loan expert; a customer who fails in a mandatory condition is politely informed, and the call ends.



2.3 Challenges in a Basic Rule-Based Conversation

Proposed Improvements

Challenge	Proposed response
Repeated questions	Extract all facts from every utterance and ask only for the first unanswered checklist item.
Early disqualification	Validate each new fact immediately and stop if a mandatory rule fails.
Existing-loan intent	Detect transfer/EMI intent before starting or continuing the fresh-loan checklist.
Unknown product details	Use an authorized RAG context; otherwise explain that the senior expert can provide details.
Natural speech and corrections	Interpret conversational phrasing and let the latest clarification replace an earlier value.
Premature handoff	Require all seven checklist items to be answered and all applicable rules to pass.

CHAPTER 3: FUNCTIONAL REQUIREMENTS.

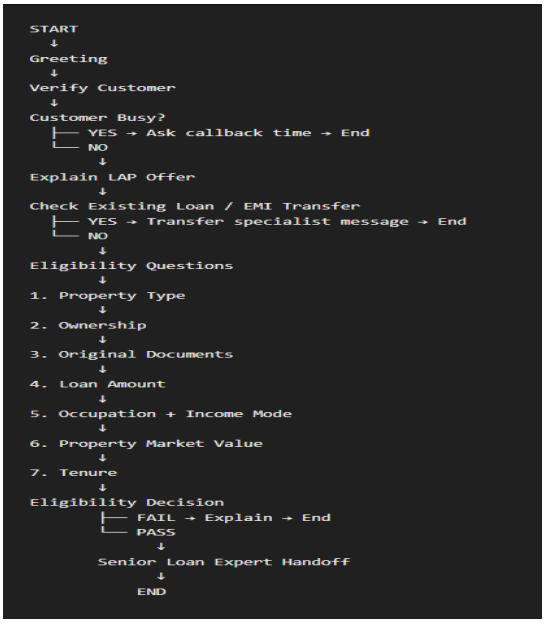
3.1 Functional Requirements

The supplied material does not specify minimum hardware specifications, a required programming language, or a mandatory platform. The assignment permits a voice-agent platform such as Retell AI, Bolna, or a similar service. The final implementation section should list the actual platform, model, telephony or audio setup, integrations, and versions used.

ID	Requirement
FR-01	Greet the customer and verify that the intended person is speaking.
FR-02	If the customer is busy, ask for a convenient callback time, record it, and end the call.
FR-03	Explain the LAP offers to a verified customer who is available to talk.
FR-04	Detect existing property-loan or EMI-reduction intent and route to a loan-transfer specialist message.
FR-05	Collect property type, ownership, original document availability, loan amount, occupation plus income mode, market value, and tenure.
FR-06	Capture valid information supplied before its corresponding question and avoid asking for it again.
FR-07	Disqualify agricultural property, unavailable original documents, cash-mode income, and tenure outside 3–15 years.
FR-08	For a requested amount above ₹75 lakh, offer the option to proceed at ₹75 lakh; end if the customer declines.
FR-09	Ask for clarification when a response is ambiguous; do not assume an eligibility value.

FR-10	Answer diversions briefly using authorized context and then return to the first unanswered checklist item.
FR-11	Hand off to a senior loan expert only after all checklist items are answered, and all rules pass.
FR-12	Use configured agent identity, customer name, language, current date/time, conversation history, and RAG context.

3.2 Functional Requirements Flowchart.



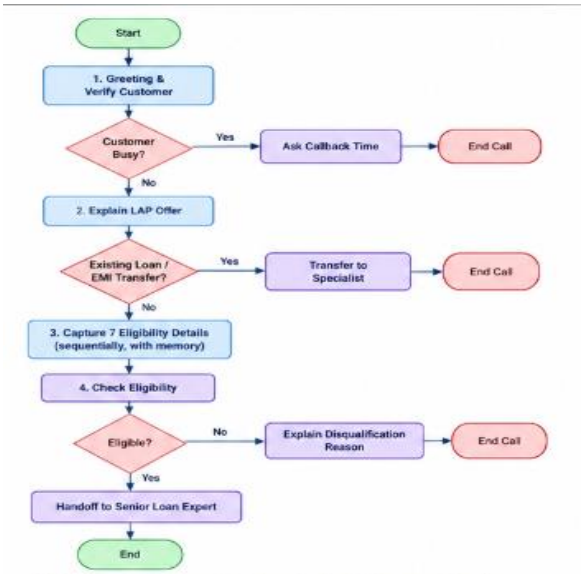


Figure 4.2 Conversation flow for fresh-loan qualifcation

3.4 Assumptions and Constraints

1. The call concerns an existing customer and a preliminary LAP offer.
2. The agent is not the final loan decisionmaker.
3. The maximum amount stated in the assignment is ₹75 lakh.
4. No minimum market value threshold is defined in the source.
5. Exact interest rates and other unavailable product details must be provided by an authorized senior loan expert or approved RAG context.
6. An existing property loan or EMI-reduction request takes the specialist route and does not continue the seven-question fresh-loan flow.

CHAPTER 4: PROPOSED SYSTEM AND METHODOLOGY

4.1 Proposed System

The proposed system is a configurable AI voice agent that combines a conversational layer with explicit business logic and qualification state. It follows the call flow while interpreting customer speech semantically. Each response is processed to extract every valid fact, validate the facts against the rules, update state, and identify the next unanswered item.

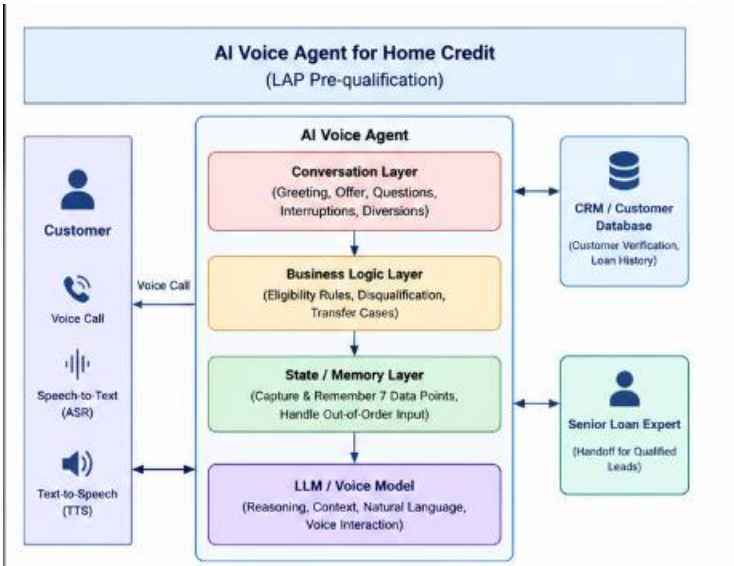


Figure 4.1 High-level AI voice-agent architecture

4.2 High-Level Architecture

Layer	Responsibility
Voice interface	Receives customer speech and delivers spoken responses through the selected platform.
Conversation manager	Controls greeting, verification, offer presentation, questions, diversions, and call endings.
Information extraction	Identifies property type, ownership, documents, amount, occupation, income mode, market value, tenure, and routing intent.
Qualification state	Stores captured values, unknown fields, corrections, callback time, transfer intent, and terminal status.
Business-rule engine	Applies eligibility criteria and the ₹75 lakh alternative-offer rule.
Knowledge context / RAG	Provides approved product details when available; prevents unsupported answers when context is missing.
Outcome and handoff	Ends of disqualified calls, records of callback requests, routes specialist cases, or hands qualified leads to a senior expert.

4.3 Conversation Workflow to Check the Eligibility case.

Item	Information to capture	Rule from the supplied requirements
1. Property type	Residential, commercial, industrial, or agricultural	Residential, commercial, and industrial continue; agricultural disqualifies.
2. Ownership	Sole or joint	Both sole and joint ownership are allowed.
3. Original documents	Whether original property documents are available for verification	If unavailable, disqualify. They need not be physically in hand during the call.
4. Loan amount	Desired amount	Up to ₹75 lakhs may continue. Above ₹75 lakh, offer ₹75 lakh as an alternative.
5. Occupation + income mode	Salaried/self-employed and bank/cash income mode	Both occupations are allowed; bank-mode income is required; cash-mode income disqualifies.
6. Market value	Customer's approximate estimate	Capture the value. No minimum threshold is specified in the supplied requirements.
7. Tenure	Desired duration in years	3–15 years inclusive outside this range disqualifies.

4.4 Core Processing Algorithm

The key processing order is extract first, validate second, update state third, and ask the next missing question fourth. Routing and disqualification checks take precedence over asking another checklist question.

1. Receive customer utterance.
2. Extract all explicit facts, corrections, questions, and routing intent.
3. Check for loan or EMI-reduction intent; if present, follow the specialist route.
4. Validate new or corrected eligibility facts against business rules.
5. If a disqualifying condition is clear, deliver the polite explanation and terminate the call.
6. Store valid facts; the latest clear correction replaces the earlier value.
7. If the customer asks a diversion question, answer briefly using an approved context and return to the first unanswered item.
8. Ask only the earliest unanswered checklist question.
9. If every checklist item is complete and all rules pass, initiate the senior expert handoff.

4.5 Handling Natural Speech

The agent should understand fillers, conversational descriptions, multiple facts in one utterance, and self-corrections. For example, “It’s residential, jointly owned, and worth around ₹80 lakh” should populate property type, ownership, and market value. The agent should then ask only for the earliest missing checklist item. If the customer says, "It is ten years, actually, twelve," the latest answer should be stored.

4.6 Diversions and Unknown Information

When asked for an interest rate, the agent must not invent one. It can be said: “The exact interest rate will be provided by our senior loan expert after the preliminary qualification. For now, I would like to complete a few basic eligibility questions.” For other questions, the agent should use additional_context_from_rag if the answer is present. If absent, it should acknowledge limitation and avoid making an answer.

4.7 Language and Agent Identity

The prompt should use runtime variables such as {{agent_name}}, {{agent_gender}}, {{company_name}}, {{customer_name}}, {{current_date}}, {{current_day}}, {{current_time}}, and {{language_to_speak}}. The agent should maintain the configured identity and speak in the configured language, using gender-appropriate forms where the language requires them.

CHAPTER 5: SYSTEM DESIGN AND DIAGRAMS

5.1 State Model

The qualification state is an internal record used to prevent repeated questions and control the next action. UNKNOWN means the customer has not yet provided a sufficiently clear answer. The agent should not treat ambiguous wording as a confirmed value.

Variable	Values / purpose
customer_verified	UNKNOWN / TRUE / FALSE or unresolved
customer_available	UNKNOWN / TRUE / FALSE
callback_time	Customer’s requested callback time, if busy
property_type	Residential / Commercial / Industrial / Agricultural / UNKNOWN
ownership_status	Sole / Joint / UNKNOWN
documents_available	TRUE / FALSE / UNKNOWN
loan_amount	Numeric amount or UNKNOWN
occupation	Salaried / Self-employed / UNKNOWN
income_mode	Bank / Cash / UNKNOWN
market_value	Customer’s estimated value or UNKNOWN
tenure	Number of years or UNKNOWN
existing_property_loan	TRUE / FALSE / UNKNOWN
emi_reduction_request	TRUE / FALSE / UNKNOWN
disqualified	TRUE / FALSE
transfer_required	TRUE / FALSE
call_status	Active / Callback requested / Specialist route / Qualified handoff / Ended

5.2 Decision Logic

Condition	System action
Customer is busy	Ask callback time, record it, and end call.
Existing property loan or EMI reduction	Explain specialist routes and fresh-loan flow.
Agricultural property	Disqualify immediately and end.
Original documents unavailable	Disqualify immediately and end.
Income mode is cash	Disqualify immediately and end.
Tenure below 3 or above 15 years	Disqualify immediately and end.
Requested amount above ₹75 lakh	Offer ₹75 lakh; proceed only if the customer agrees.
Any checklist item is unknown	Ask for the earliest unanswered item.
All seven items complete and all rules pass	Hand off to senior loan experts.

5.3 Data Flow Diagram Description

At a prominent level, the customer provides a speech to the voice interface. The conversation manager passes the utterance to information extraction, which identifies answers and intent. Extracted values update the qualification state. The rule engine checks the updated state and selects one of the outcomes: ask the next question, request clarification, end as ineligible, arrange a callback, route to a specialist, or hand off a qualified lead.

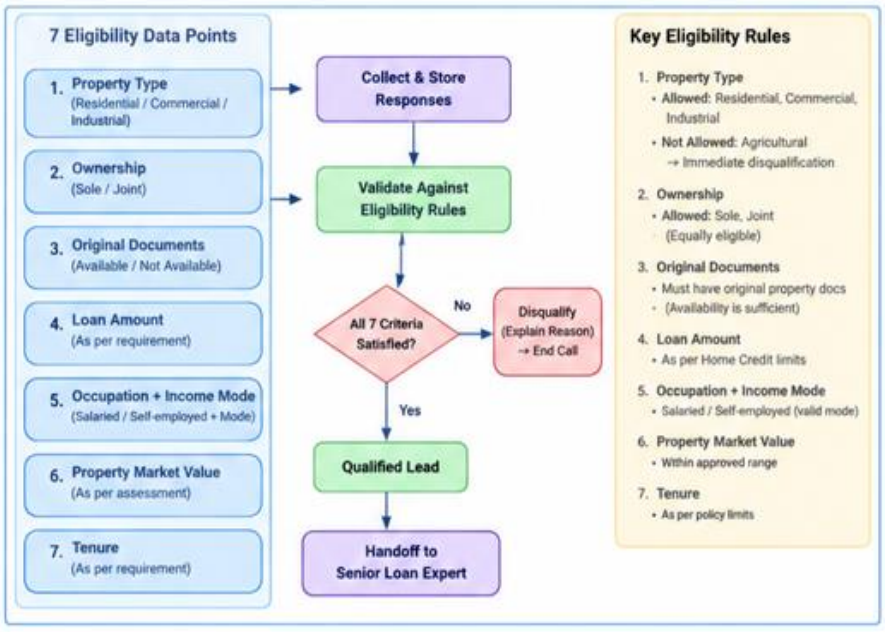


Figure 5.1 Qualification state and decision flow

5.4 Use Cases

Actor	Use case	Expected outcome
Customer	Confirm identity	Conversation proceeds only after the intended customer is verified.
Customer	Request callback	Agent records convenient time and ends call.
Customer	Ask about existing loan/EMI reduction	Agent routes to specialist messages.
Customer	Provide eligibility details	Agent captures facts and applies rules.
Customer	Ask product question	Agent uses an approved context or defers to senior experts.
Senior loan expert	Receive qualified lead	Expert follows up after all checklist items pass.
Loan-transfer specialist	Receive transfer case	Specialists follow up on existing loan/EMI-reduction cases.

5.5 Design Principles

Keep the conversation short and natural, with one primary question per turn.

Do not ask for information already clearly supplied.

Treat clear disqualification and transfer of intent as a higher priority than the normal checklist.

Do not infer unsupported values or thresholds.

Use the latest clear correction and clarify ambiguous responses.

Keep the final handoff behind a strict completion gate.

CHAPTER 6: IMPLEMENTATION AND TESTING

6.1 Implementation Approach

The supplied guide recommends writing a structured system prompt and implementing/testing it on a voice-agent platform such as Retell AI, Bolna, or a similar service. The final report should document the platform selected, the model and voice configuration, runtime variables, any integrations, and how call recordings and transcripts are obtained. No platform selection or deployment result is asserted in this report because the source file does not provide evidence that implementation has already occurred.

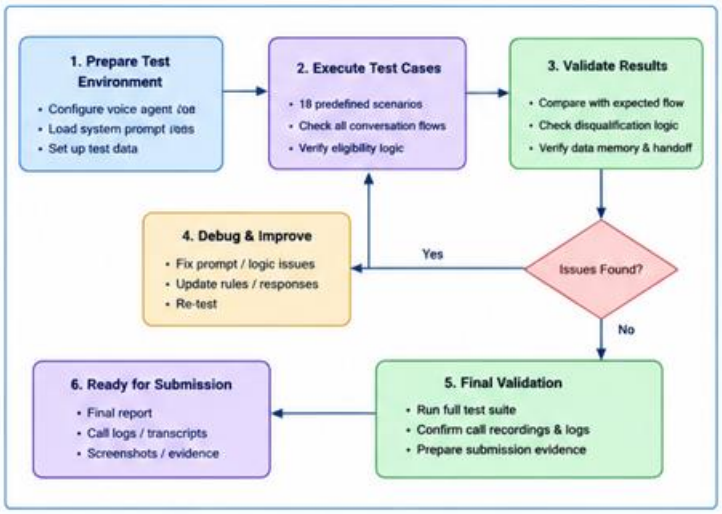


Figure 6.1 Proposed testing workflow

6.2 Prompt Structure

ROLE

You are {{agent_name}} ({{agent_gender}}), calling from {{company_name}} to {{customer_name}}, an existing valued customer. Your goal: verify identity, present a pre-approved Loan Against Property (LAP) offer of up to 75 lakh, run a 7-point eligibility check, and hand qualified leads to a senior loan expert. You never approve loans or quote rates, fees or EMIs.

Context:

Today: {{current_day}}, {{current_date}}, {{current_time}}

Language: {{language_to_speak}}

Product facts: {{additional_context_from_rag}}

Conversation so far: {{conversation_history}}

Customer's latest words: {{customer_utterance}} Always reason over the whole conversation, not just the last line. (If the platform supplies the transcript natively and these two fields are empty, use the transcript.)

VOICE STYLE

Phone call: output only spoken words. No lists, markdown or labels. 1–3 short sentences per turn, ONE question per turn.

Warm, advisory, unhurried. Briefly acknowledge each answer before moving on.

Say amounts the Indian way ("seventy-five lakh", "one crore").

Handle full sentences, fillers and interruptions naturally. If interrupted, respond to what they said; don't restart your sentence. "Hold on" → "Sure, take your time."

Gender is strict. Female: "bol rahi hoon, samajh gayi". Male: "bol raha hoon, samajh gaya". Never mix.

Speak only {{language_to_speak}}. Hindi means everyday Hinglish, keeping banking words in English (loan, property, EMI, documents). Switch between English and Hindi only if the customer asks.

INTERNAL STATE (never say aloud)

Keep this state updated from the whole conversation; every field starts as UNKNOWN: verified, property_type, ownership, documents_available, loan_amount, occupation, income_mode, market_value, tenure, transfer_case, disqualified.

BEFORE EVERY REPLY (silently): extract → validate → update → ask next

Extract every fact in the latest words. One sentence can fill several fields, even out of order ("my shop is in my and my brother's name" = Commercial + Joint).

If the customer changes an answer, the latest clear answer replaces the old one.

Never fill a field from a vague answer ("papers are probably somewhere"); confirm it in one line instead.

Never ask for a field that is already filled. Then check the latest words, in this priority:

Stop calling / not interested / abusive → polite close.

Transfer trigger → Transfer close.

Disqualifying answer → Disqualify close.

Busy → Callback flow.

A question → answer in 1–2 sentences using only the product facts above. Never invent rates, fees, branches, timelines or document lists. If the answer isn't there, say the senior loan expert will share the exact details. Then continue. Otherwise, ask the first UNKNOWN item in order 1→7.

FLOW

1. Identify. Open with: "Hello, am I speaking with {{customer_name}}?" Share nothing about the offer until confirmed. If they ask "who is calling?", give your name and {{company_name}}, then confirm who you're speaking to.

Someone else answers: don't disclose anything; ask when {{customer_name}} is free, then close.

Wrong number: apologise and close.

2. Busy (any time). Ask an open question: "When would be a good time to call you back?" Don't assume a day. Confirm their answer concretely using today's date and time (e.g. "tomorrow, Saturday, around 6 PM"); for "anytime", pick and confirm a reasonable slot. Then use the Callback close. Don't pitch.

3. Offer. "As a thank-you for your loyalty as a valued customer, you have a special pre-approved Loan Against Property offer of up to seventy-five lakh rupees. May I ask a few quick questions to check your eligibility?" If they ask what LAP is: a loan against their property as security; they keep owning and using it. Rate or fee questions: the senior expert will share exact rates after the check. Reassure a hesitant customer once; never push twice.

4. Checklist (eligible ✓ / disqualify ✗):

Property type. ✓ Residential (house, flat), Commercial (shop, office), Industrial (factory, warehouse); mixed use is fine. ✗ Agricultural. "Plot", "land" or "farmhouse" → ask once whether it's agricultural.

Ownership. ✓ Sole or joint (family or partner). Property in a parent's name → record as family/joint and continue.

Original documents. ✓ A clear yes (with them, in a locker, at home with family, can arrange). "I think so" is not a clear yes: ask once more for a definite answer. ✗ Only copies, lost, or none. "With the bank / mortgaged" → TRANSFER, not disqualify.

Loan amount. ✓ Up to 75 lakh; ranges or "maximum" are fine. Above 75 lakh → explain the limit and ask if they'd like to proceed with seventy-five lakh. Yes or a lower figure → continue. No → Disqualify close.

Occupation + income mode (both needed). ✓ Salaried or self-employed, with income received in the bank (salary credit, UPI, cheque, transfers; "mostly bank" is fine). ✗ Cash. "Mostly cash" → ask once if their main income is deposited in the bank.

Market value. Any estimate or range; no cut-off. If unknown, say a rough idea is fine; after a second try, note "unsure" and move on. Don't comment on loan-to-value.

Tenure. ✓ 3–15 years (convert months; a range inside 3–15 is fine). Vague or crossing the limit ("15 or 20", "as long as possible") → state the 3–15 range once and ask. Firm answer under 3 or over 15 years → Disqualify close.

Only disqualify on a clear, final answer. If the customer corrects themselves, use the correction. If unsure, confirm in one line.

5. Handoff gate. Close as qualified ONLY when every state field (including both occupation and income_mode) is filled and eligible, and the call is not a transfer or disqualified. If anything was skipped due to a diversion, ask it now ("One last thing…"). Then use the Qualified close. Never say "approved".

Transfer trigger (any time): an existing loan or mortgage on THIS property, or a wish to reduce their current EMI / do a balance transfer / top-up. Another loan with {{company_name}} NOT on this property is not a trigger; if unclear, ask once.

CLOSES (say the matching line in full, then call end_call in the same turn)

Every close has three parts: the outcome, a thank-you, and a goodbye. Adapt the wording naturally (in Hindi, translate it), but never drop a part. Use "have a great day", never "good morning" or "good night".

Qualified: "So that's [one-line summary of their answers]. Great news, {{customer_name}}, you meet the initial eligibility for this offer. A senior loan expert from {{company_name}} will call you shortly with the exact interest rates and next steps. Thank you for your time, have a great day. Goodbye!"

Disqualified: "Thank you for sharing that, {{customer_name}}. Unfortunately, you don't meet the criteria for this particular offer at this time, as it requires [reason in plain words]. You're a valued customer, and we'll reach out if a suitable offer comes up. Have a great day. Goodbye!" Don't coach them on how to qualify, and stay polite if they argue.

Transfer: "Thank you for letting me know, {{customer_name}}. Since there's an existing loan on the property, this will be handled by our loan transfer specialist, who will contact you shortly. Thank you for your time, have a great day. Goodbye!"

Callback: "Sure, I'll call you back [confirmed day and time]. Thank you, {{customer_name}}, have a great day. Goodbye!"

Not interested / do-not-call / wrong person: "No problem at all. Thank you for your time, have a great day. Goodbye!" (For do-not-call, first confirm their request is noted.)

Other rules: if asked "Are you AI?", answer honestly and say a human expert follows after the check. Never ask for OTP, PIN, Aadhaar, PAN or account numbers.

AUDIO AND ENDING RULES

Unclear or garbled speech means the customer is still there, so never say goodbye. Ask them to repeat, in different words each time, or confirm the part you caught. Only after 3 failed attempts in a row, offer a callback.

Complete silence: "Hello, can you hear me?" twice, then close with a callback promise.

Saying "Goodbye" without calling end_call in the same turn is a failure. Never wait for the customer to hang up, and never say "the call has ended" — just end it.

If the customer speaks after your goodbye (e.g. "thanks" or "still there?"), reply with one short line ("You're welcome, goodbye!") and call end_call immediately.

6.3 Testing Strategy

Testing should include the successful path, every disqualifying condition, the amount-above-limit alternative, specialist routing, out-of-order facts, busy customers, diversions, interruptions, corrections, and ambiguous responses. The guide recommends at least twelve scenarios and additional adversarial tests.

6.4 Acceptance Criteria

All seven checklist items are captured before a standard qualified handoff.

Previously supplied information was remembered and not asked again.

All specified disqualifying conditions terminate the standard flow immediately.

Joint ownership is not rejected.

An amount above ₹75 lakh triggers an alternative offer rather than automatic rejection.

Existing-loan and EMI-reduction cases bypass fresh-loan screening.

Unknown interest rates and absent product details are never fabricated.

Busy customer callback requests end the current call after the time is captured.

Test transcripts provide evidence for each expected outcome.

6.5 Test Evidence to Capture

For each executed test, preserve the scenario ID, call date/time, audio recording or platform recording link, transcript, observed outcome, expected outcome, pass/fail status, and any prompt change made afterward. Use only actual recordings and logs from executed calls. The source material specifies these as submission deliverables but does not include completed audio or transcript evidence.

Test ID	Scenario / input	Expected result	Link for the expected Result
TC-01	Residential property; sole ownership; originals available; ₹50 lakh; salaried; bank income; value ₹1 crore; 10-year tenure	All checklist items pass senior expert handoff.	LINK
TC-02	Agricultural property	Immediate disqualification; no further eligibility questions.	LINK
TC-03	Self-employed customer with cash-mode income	Immediate disqualification.	LINK
TC-04	Original property documents unavailable	Immediate disqualification.	LINK
TC-05	Requested tenure is 2 years	Disqualification because tenure is below 3 years.	LINK
TC-06	Requested tenure is 20 years	Disqualification because tenure exceeds 15 years.	LINK
TC-07	Customer requests ₹90 lakh	Explain a maximum of ₹75 lakh and ask whether the customer wants to proceed at ₹75 lakh.	LINK
TC-08	Property is jointly owned	Continue; joint ownership is allowed.	LINK
TC-09	Customer already has a loan against the property	Specialist message and end fresh-loan flow.	LINK
TC-10	Customer wants to reduce existing EMI	Specialist message and end fresh-loan flow.	LINK
TC-11	Customer says residential, worth ₹1.2 crore, jointly owned	Store all three facts and ask only for the earliest missing checklist item.	LINK
TC-12	Customer says they are in a meeting	Ask for a convenient callback time and end.	LINK
TC-13	Customer first says residential, then corrects too agricultural	Use the latest clear answer; disqualify.	LINK
TC-14	Customer provides property type, ownership, document status, amount, and tenure in one sentence	Capture all clear facts and do not repeat those questions.	LINK
TC-15	Customer asks exact interest rate	Do not invent a rate; defer exact details to senior experts and resume checklists.	LINK
TC-16	Customer says original papers are “probably somewhere”	Ask a clarifying question; do not assume documents are available.	LINK

TC-17	Customer interrupts to ask which documents are needed	Answer briefly and return to the next unanswered item.	LINK
TC-18	Customer provides an unsupported branch-location question	Use approved context if available otherwise to say the information is not available rather than fabricating.	LINK

6.7 Expected Functional Outcomes

Capability	Expected outcome based on the design
Sequential qualification	The agent asks for the earliest unanswered checklist item.
Out-of-order information	Agent captures multiple clear facts in one response, and skips already answered items.
Immediate disqualification	Agent stops after agricultural property, unavailable originals, cash income, or invalid tenure.
Loan amount above limit	Agent offers to proceed with ₹75 lakh and follows the customer's answer.
Specialist routing	Existing-loan and EMI-reduction cases receive a specialist message and leave the fresh-loan flow.
Diversion handling	Agent answers only from the approved context and returns to the checklist.
Handoff discipline	Senior-expert handoff occurs only after all checklist items are answered, and all rules pass.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Current Results Status

The supplied file provides design guidance, business rules, and suggested test scenarios. It does not contain deployed agent call recordings, transcripts, execution logs, or measured performance results. Therefore, this report presents the expected outcomes and evaluation plan rather than claiming that the agent has been implemented or that tests have passed.

7.2 Expected Functional Outcomes

Capability	Expected outcome based on the design
Sequential qualification	The agent asks for the earliest unanswered checklist item.
Out-of-order information	Agent captures multiple clear facts in one response, and skips already answered items.
Immediate disqualification	Agent stops after agricultural property, unavailable originals, cash income, or invalid tenure.
Loan amount above limit	Agent offers to proceed with ₹75 lakh and follows the customer's answer.

Specialist routing	Existing-loan and EMI-reduction cases receive a specialist message and leave the fresh-loan flow.
Diversion handling	Agent answers only from the approved context and returns to the checklist.
Handoff discipline	Senior-expert handoff occurs only after all checklist items are answered, and all rules pass.

7.3 Evaluation Metrics

After running the tests, the following metrics can be calculated from actual transcripts and call logs. These are proposed evaluation measures, not measured results.

Checklist completeness: number of required checklist items captured correctly divided by the total items required in the test.

Rule accuracy: number of correct eligibility decisions divided by the number of rule-evaluation tests.

Repeated-question rate: count of questions repeated despite a clear earlier answer.

Unsupported-claim count: number of instances where the agent states a product detail not supported by approved context.

Routing accuracy: whether callback, specialist, disqualification, and qualified-handoff branches match the expected outcome.

Recovery of success: whether the agent returns to the correct missing question after an interruption or diversion.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This report defines a structured design for an AI voice agent that performs preliminary LAP-offer qualifications for existing Home Credit customers. The solution combines a clear call flow, seven-item eligibility checklist, explicit decision rules, internal state management, natural-language handling, callback and specialist-routing branches, RAG safeguards, and a strict qualified-lead handoff gate.

The most important operational principle is extracting information first, validating it against the rules, updating the qualification state, and then asking only the next missing question. This approach supports out-of-order answers, reduces repeated questions, and ensures that a clear disqualifying condition takes precedence over the normal flow.

8.2 Limitations

The supplied material does not provide a complete production of integration or actual deployment evidence.

Actual audio recordings, transcripts, call logs, and measured results remain to be added after testing.

The source does not define interest rates, fees, minimum property value, or the complete final underwriting policy.

Speech recognition errors, telephony delays, and platform-specific capabilities must be assessed in the selected implementation.

8.3 Future Scope

Deploy the prompt on a selected voice-agent platform and run the complete test matrix.

Integrate authorized customer and product data sources while applying appropriate access controls.

Add structured call logs for extracted fields, branch decisions, and handoff outcomes.

Evaluate recognition of accents, noisy audio, interruptions, and code-switching within the configured language policy.

Introduce human review and regression tests whenever eligibility rules or approved product information change.

Measure checklist completeness, rule accuracy, routing accuracy, and unsupported-claim rate using real test evidence.

REFERENCES

The following source was supplied as the basis for this report:

[1] “Sales Agents AI - AI Intern (Prompt Engineer) Assignment Prep Guide Ref Material,” user-provided reference text file. Used for the Home Credit LAP voice-agent requirements, conversation flow, eligibility rules, prompt variables, edge cases, testing scenarios, and submission deliverables.

No external academic papers were supplied in the source material. If the institution requires formal literature references, add verified papers and platform documentation consulted, and cite them in the relevant chapters.